

Raúl Figueroa Sierra

Personal Information

Full name: Raúl Figueroa Sierra
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Research Interests

Set theory, with emphasis on descriptive set theory, infinite combinatorics, forcing, cardinal invariants of the continuum, and set-theoretic topology.

Education

Universidad de los Andes

Bogotá, Colombia

- Ph.D. in Mathematics (2019–2025)
Thesis: *Combinatorics of ideals on ω : the Category Dichotomy and the Measure Dichotomy*.
Advisors: Michael Hrušák (UNAM) and Carlos Di Prisco (Uniandes).
- M.Sc. in Mathematics (2017–2019)
Thesis: *Cardinal Quantifiers in Continuous Logic*.
Advisor: Xavier Caicedo Ferrer.

Universidad Nacional de Colombia

Bogotá, Colombia

- B.Sc. in Mathematics (2012–2017)
Thesis: *On Some Categorical Notions of Non-Elementary Classes*.
Advisor: Pedro Zambrano Ramírez.

Publications and preprints

- *The Category Dichotomy for Ideals*.
Joint with Alan Dow, Osvaldo Guzmán, and Michael Hrušák.
Annals of Pure and Applied Logic (2026).
- *Are scales Fréchet?*
Joint with Osvaldo Guzmán, Michael Hrušák, and Adam Kwela.
Preprint, [arXiv:2603.15567v2](https://arxiv.org/abs/2603.15567v2) (2026).
- *Compactness of continuous logic with generalized quantifiers*.
Joint with Xavier Caicedo.
In preparation.
- *Combinatorics of the Arkhangel'skii–Franklin space S_ω* .
In preparation.

Selected Talks

- *The optimal projective complexity of a counterexample to the Category Dichotomy.*
XXI Latin American Symposium on Mathematical Logic (SLALM), Bogotá, Colombia, June 1–5, 2026.
- *Ideales de subconjuntos densos en ninguna parte sobre espacios de Dow.*
XIII Simposio Nororiental de Matemáticas, Bucaramanga, Colombia, December 1–5, 2025.
- *Espacios topológicos y la Dicotomía de Categoría.*
Taller Mexicano de Topología, Mexico City, Mexico, June 15–19, 2025.
- *Pathological Lower Semicontinuous Submeasures and F_σ Ideals on ω* (Poster).
Young Set Theory Workshop, Münster, Germany, May 29 – June 3, 2023.
- *Compact Extensions of Continuous Logic and Partition Properties.*
XVIII Latin American Symposium on Mathematical Logic (SLALM), Concepción, Chile, 2019.

Teaching Experience

Universidad de los Andes, Departamento de Matemáticas

- **Adjunct Lecturer (Profesor de Cátedra)** 2025-II
 - Teaching load: 12 hours/week.
 - Courses: Linear Algebra I, Calculus III, Precalculus.
 - Responsibilities: course planning, lectures, exam preparation, grading, student advising.
- **Graduate Teaching Assistant** 2017-II – 2025-I
 - Teaching load: 6 hours/week.
 - Courses assisted: Linear Algebra, Differential Calculus, Integral Calculus and Differential Equations, Linear Algebra and Calculus III, Integral Calculus and Probability, Multivariable Calculus.
 - Duties: problem sessions, grading, academic support.
- **Research Assistant** June–July 2018, June–July 2019

Universidad Nacional de Colombia, Facultad de Ciencias

- **Teaching Assistant** 2014–2017
 - Basic Mathematics Program (March–June 2016; June–July 2016).
 - Faculty of Sciences (September–December 2014; February–June 2015).
 - Youth summer courses on mathematical outreach (March–May 2017).

Last updated: May 22, 2026